

```
*****
*
*      STAAD.Pro
*      Version 2007   Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=   JAN 18, 2013
*      Time=   15: 1: 8
*
*      USER ID:
*****

1. STAAD SPACE
INPUT FILE: 20121120 T-Goal Post 029-031 FR.STD
2. START JOB INFORMATION
3. ENGINEER DATE 12-MAY-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 2 5 1 -5.73; 3 5 1.74 -4.93; 4 5 1.74 -5.73; 6 5 1.74 -6.53
9. MEMBER INCIDENCES
10. 1 4 3; 3 2 4; 5 6 4
11. DEFINE MATERIAL START
12. ISOTROPIC STEEL
13. E 2.05E+008
14. POISSON 0.3
15. DENSITY 76.8195
16. ALPHA 1.2E-005
17. DAMP 0.03
18. END DEFINE MATERIAL
19. MEMBER PROPERTY JAPANESE
20. 1 5 TABLE ST H200X200X8
21. 3 TABLE ST H200X200X8
22. CONSTANTS
23. BETA 90 MEMB 3
24. MATERIAL STEEL ALL
25. SUPPORTS
26. 2 FIXED
27. ***** DEAD LOAD *****
28. LOAD 1 LOADTYPE NONE TITLE DL
29. SELFWEIGHT Y -1
30. JOINT LOAD
31. 4 FY -200
32. ***** FRICTIONAL LOAD *****
33. LOAD 2 LOADTYPE NONE TITLE FL
34. JOINT LOAD
35. 4 FX -23
36. ***** WIND LOAD X-DIRECTION*****
37. LOAD 3 LOADTYPE NONE TITLE WLX
38. MEMBER LOAD
39. 1 3 5 UNI GX -0.255
40. ***** WIND LOAD Z-DIRECTION*****
41. LOAD 4 LOADTYPE NONE TITLE WLZ
42. MEMBER LOAD
43. 3 UNI GZ -0.255
44. ***** NOTIONAL LOAD X-DIRECTION*****
45. LOAD 5 LOADTYPE NONE TITLE NLX
46. SELFWEIGHT X -0.015
47. JOINT LOAD
48. 4 FX -3
49. *****NOTIONAL LOAD Z-DIRECTION*****
50. LOAD 6 LOADTYPE NONE TITLE NLZ
51. SELFWEIGHT Z -0.015
52. JOINT LOAD
53. 4 FZ -3
```

```
54. ***** LOAD COMBINATION - SERVICE *****
55. LOAD COMB 7 1.0DL+1.0FL
56. 1 1.0 2 1.0
57. LOAD COMB 8 1.0DL+1.0FL+1.0WLX
58. 1 1.0 2 1.0 3 1.0
59. LOAD COMB 9 1.0DL+1.0FL+1.0WLZ
60. 1 1.0 2 1.0 4 1.0
61. LOAD COMB 10 1.0DL+1.0FL+1.0NLX
62. 1 1.0 2 1.0 5 1.0
63. LOAD COMB 11 1.0DL+1.0FL+1.0NLZ
64. 1 1.0 2 1.0 6 1.0
65. ***** LOAD COMBINATION - ULTIMATE *****
66. LOAD COMB 12 1.0DL+1.4FL
67. 1 1.0 2 1.4
68. LOAD COMB 13 1.4DL+1.4FL
69. 1 1.4 2 1.4
70. LOAD COMB 14 1.0DL+1.4FL+1.4WLX
71. 1 1.0 2 1.4 3 1.4
72. LOAD COMB 15 1.0DL+1.4FL+1.4WLZ
73. 1 1.0 2 1.4 4 1.4
74. LOAD COMB 16 1.0DL+1.4FL+1.4NLX
75. 1 1.0 2 1.4 5 1.4
76. LOAD COMB 17 1.0DL+1.4FL+1.4NLZ
77. 1 1.0 2 1.4 6 1.4
78. PERFORM ANALYSIS

P R O B L E M   S T A T I S T I C S
-----

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =      4/      3/      1

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES =      6, TOTAL DEGREES OF FREEDOM =      18

79. LOAD LIST 7 TO 11
80. PRINT JOINT DISPLACEMENTS ALL

JOINT DISPLACEMENT (CM   RADIANS)   STRUCTURE TYPE = SPACE
-----

JOINT   LOAD   X-TRANS   Y-TRANS   Z-TRANS   X-ROTAN   Y-ROTAN   Z-ROTAN

2       7       0.0000   0.0000   0.0000   0.0000   0.0000   0.0000
      8       0.0000   0.0000   0.0000   0.0000   0.0000   0.0000
      9       0.0000   0.0000   0.0000   0.0000   0.0000   0.0000
     10       0.0000   0.0000   0.0000   0.0000   0.0000   0.0000
     11       0.0000   0.0000   0.0000   0.0000   0.0000   0.0000
3       7      -0.1015  -0.0118   0.0000   0.0000   0.0000   0.0019
      8      -0.1040  -0.0118   0.0000   0.0000   0.0000   0.0020
      9      -0.1015  -0.0117  -0.0002   0.0000   0.0000   0.0019
     10      -0.1148  -0.0118   0.0000   0.0000   0.0000   0.0022
     11      -0.1015  -0.0050  -0.0060  -0.0001   0.0000   0.0019
4       7      -0.1015  -0.0114   0.0000   0.0000   0.0000   0.0019
      8      -0.1036  -0.0114   0.0000   0.0000   0.0000   0.0020
      9      -0.1015  -0.0114  -0.0002   0.0000   0.0000   0.0019
     10      -0.1148  -0.0114   0.0000   0.0000   0.0000   0.0022
     11      -0.1015  -0.0114  -0.0060  -0.0001   0.0000   0.0019
5       7      -0.1015  -0.0118   0.0000   0.0000   0.0000   0.0019
      8      -0.1040  -0.0118   0.0000   0.0000   0.0000   0.0020
      9      -0.1015  -0.0119  -0.0002   0.0000   0.0000   0.0019
     10      -0.1148  -0.0118   0.0000   0.0000   0.0000   0.0022
```

Friday, January 18, 2013, 03:01 PM

***** END OF LATEST ANALYSIS RESULT *****

81. PRINT SUPPORT REACTION ALL

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
2	7	23.00	201.14	0.00	0.00	0.00	-17.02
	8	23.60	201.14	0.00	0.00	0.00	-17.39
	9	23.00	201.14	0.19	0.07	0.00	-17.02
	10	26.02	201.14	0.00	0.00	0.00	-19.25
	11	23.00	201.14	3.02	2.23	0.00	-17.02

***** END OF LATEST ANALYSIS RESULT *****

82. LOAD LIST 12 TO 17
83. PRINT MEMBER FORCES ALL

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)								
MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
1	12	4	0.00	0.39	0.00	0.00	0.00	0.16
		3	0.00	0.00	0.00	0.00	0.00	0.00
	13	4	0.00	0.55	0.00	0.00	0.00	0.22
		3	0.00	0.00	0.00	0.00	0.00	0.00
	14	4	0.00	0.39	-0.29	0.00	0.11	0.16
		3	0.00	0.00	0.00	0.00	0.00	0.00
	15	4	0.00	0.39	0.00	0.00	0.00	0.16
		3	0.00	0.00	0.00	0.00	0.00	0.00
	16	4	0.00	0.39	-0.01	0.00	0.00	0.16
		3	0.00	0.00	0.00	0.00	0.00	0.00
17	4	0.01	0.39	0.00	0.00	0.00	0.16	
	3	0.00	0.00	0.00	0.00	0.00	0.00	
3	12	2	201.14	0.00	32.20	0.00	-23.83	0.00
		4	-200.78	0.00	-32.20	0.00	0.00	0.00
	13	2	281.60	0.00	32.20	0.00	-23.83	0.00
		4	-281.09	0.00	-32.20	0.00	0.00	0.00
14	2	201.14	0.00	33.04	0.00	-24.35	0.00	
	4	-200.78	0.00	-32.77	0.00	0.00	0.00	
15	2	201.14	0.26	32.20	0.00	-23.83	0.10	
	4	-200.78	0.00	-32.20	0.00	0.00	0.00	
16	2	201.14	0.00	36.42	0.00	-26.95	0.00	
	4	-200.78	0.00	-36.42	0.00	0.00	0.00	
17	2	201.14	4.22	32.20	0.00	-23.83	3.12	
	4	-200.78	-4.22	-32.20	0.00	0.00	0.00	
5	12	6	0.00	0.00	0.00	0.00	0.00	0.00
		4	0.00	0.39	0.00	0.00	0.00	-0.16
	13	6	0.00	0.00	0.00	0.00	0.00	0.00
		4	0.00	0.55	0.00	0.00	0.00	-0.22
14	6	0.00	0.00	0.00	0.00	0.00	0.00	

Friday, January 18, 2013, 03:01 PM

15	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	0.39	0.00	0.00	0.00	-0.16
16	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	0.39	-0.01	0.00	0.00	-0.16
17	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.01	0.39	0.00	0.00	0.00	-0.16

***** END OF LATEST ANALYSIS RESULT *****

84. LOAD LIST ALL
85. PARAMETER 1
86. CODE BS5950
87. BEAM 2 ALL
88. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1 ST	H200X200X8	PASS	BS-4.8.3.3.2	0.003	14
		0.00	0.11	0.16	-
3 ST	H200X200X8	PASS	BS-4.8.3.2	0.408	16
		201.14 C	-26.95	0.00	0.00
5 ST	H200X200X8	PASS	BS-4.9	0.003	14
		0.00	0.11	0.16	0.80

***** END OF TABULATED RESULT OF DESIGN *****

89. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= JAN 18,2013 TIME= 15: 1: 9 ****

* For questions on STAAD.Pro, please contact *

* Research Engineers Offices at the following locations *

* * *

* Telephone Email *

* USA: +1 (714)974-2500 support@bentley.com *

* CANADA +1 (905)632-4771 detech@odandetech.com *

* UK +44(1454)207-000 support@bentley.com *

* NORWAY +47 67 57 21 30 staaeddr.no *

* SINGAPORE +65 6225-6158 support@bentley.com *

* INDIA +91(033)4006-2021 support@bentley.com *

* JAPAN +81(03)5952-6500 eng-eye@crc.co.jp *

* CHINA +86(411)8479-1166 support@bentley.com *

* THAILAND +66(0)2645-1018/19 support@bentley.com *

* * *

* North America support@bentley.com *

* Europe support@bentley.com *

* Asia support@bentley.com *
